

Quiz 5

✓ ① [A] Projections are invertible. F.

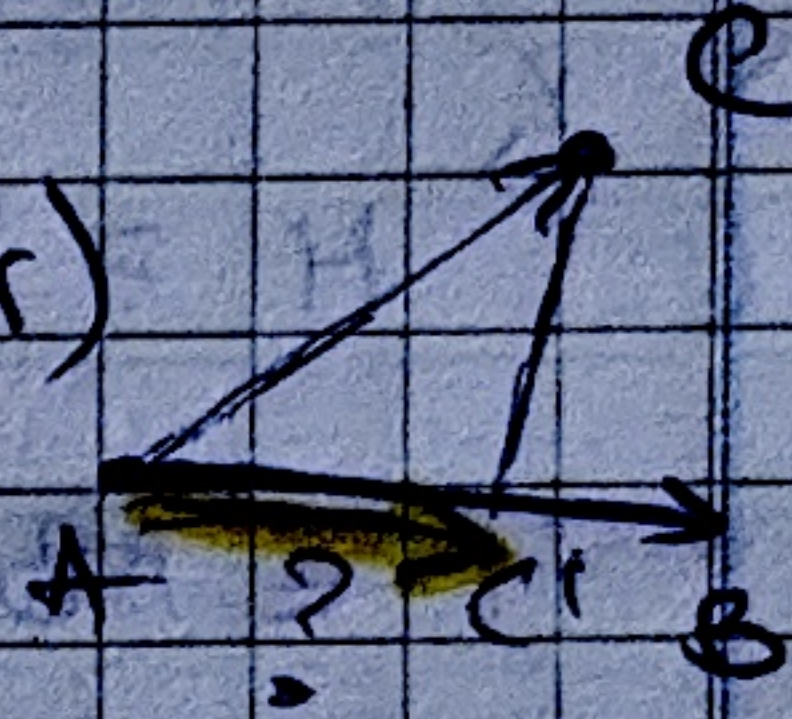
[C] Reflections are invertible. T.

✓ ③ $[A]_K = \begin{bmatrix} -5 \\ 2 \end{bmatrix}$, $[B]_K = \begin{bmatrix} -2 \\ 5 \end{bmatrix}$, $[C]_K = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

orthogonal proj. of C on AB.

$$\vec{AB} = B - A = \begin{bmatrix} 3 \\ 3 \end{bmatrix} = v \text{ (dir. vector)}$$

$$\vec{AC} = C - A = \begin{bmatrix} 6 \\ -3 \end{bmatrix}$$



(projection vector) $\vec{AC'} = \frac{\langle \vec{AC}, v \rangle}{|v|^2} \cdot v = \frac{9}{18} (3, 3) = \left(\frac{3}{2}, \frac{3}{2}\right)$.

$$\langle \vec{AC}, v \rangle = \langle (6, -3), (3, 3) \rangle = 18 - 9 = 9.$$

$$|v| = \sqrt{3^2 + 3^2} = \sqrt{2 \cdot 3^2} = 3\sqrt{2}.$$

$$|v|^2 = 9 \cdot 2 = 18$$

$$C' = A + \vec{AC'} = \begin{bmatrix} -5 \\ 2 \end{bmatrix} + \begin{bmatrix} \frac{3}{2} \\ \frac{3}{2} \end{bmatrix} = \begin{bmatrix} -\frac{7}{2} \\ \frac{7}{2} \end{bmatrix}$$

⑤ $(A, \vec{AB}, \vec{AC}) = K$

reflection in line AB || $\vec{BC}$ of the midpoint of $[BC]$.

$$K - \text{mid. of } BC \Rightarrow K \left(\frac{1+0}{2}, \frac{0+1}{2} \right) = \left(\frac{1}{2}, \frac{1}{2} \right)$$

$$\vec{AP} = x \cdot \vec{AB} + y \cdot \vec{AC}$$

$$\vec{AB} = 1 \cdot \vec{AB} + 0 \cdot \vec{AC} \Rightarrow [B]_K = (1, 0) \Rightarrow \text{x-axis.}$$

$$\vec{AC} = 0 \cdot \vec{AB} + 1 \cdot \vec{AC} \Rightarrow [C]_K = (0, 1).$$

①

AB : line of reflection : x-axis ($y=0$)

d : direction of reflection: $\parallel BC$

$$BC = C - B = (0, 1) - (1, 0) = (-1, 1).$$

• reflection of M across line AB parallel to dir. d:

$$\begin{cases} M = P + \alpha \cdot d & (1) \\ \hookrightarrow M' = P - \alpha \cdot d & (\text{ref.}) \end{cases}$$

$$P \in AB \Rightarrow P(x_P, 0).$$

dir. vector: $d(-1, 1)$.

$$(1): \left(\frac{1}{2}, \frac{1}{2}\right) = x_P, 0 + \alpha \cdot (-1, 1)$$

$$\Rightarrow \left(\frac{1}{2}, \frac{1}{2}\right) = (x_P - \alpha, \alpha)$$

$$\Rightarrow \begin{cases} \frac{1}{2} = x_P - \alpha \\ \frac{1}{2} = \alpha \end{cases} \Rightarrow \frac{1}{2} = x_P - \frac{1}{2} \Rightarrow x_P = 1 \Rightarrow P(1, 0).$$

$$M' = P - \alpha \cdot d$$

$$= (1, 0) - \frac{1}{2}(-1, 1)$$

$$= \left(1 + \frac{1}{2}, -\frac{1}{2}\right)$$

$$= \left(\frac{3}{2}, -\frac{1}{2}\right)$$

④. A, B, C, D.

✓ orthogonal projection of D on (ABC).

$$\vec{AB} = B - A = (0, 2, 0)$$

$$\vec{AC} = C - A = (1, 3, -2)$$

$$n = \vec{AB} \times \vec{AC} = \begin{vmatrix} i & j & k \\ 0 & 2 & 0 \\ 1 & 3 & -2 \end{vmatrix} = (-4, 0, -2) \sim (2, 0, 1).$$

$$(ABC): 2(x+1) + 0 \cdot (y+1) + 1 \cdot (z-1) = 0.$$

$$\Rightarrow 2x + z + 1 = 0.$$

$$\text{proj. vector: } \frac{\langle \vec{AD}, m \rangle}{|m|^2} \cdot m = \frac{3}{5} (2, 0, 1) = \left(\frac{6}{5}, 0, \frac{3}{5} \right)$$

$$\langle \vec{AD}, m \rangle = \langle (1, 2, 1), (2, 0, 1) \rangle = 2 + 1 = 3$$

$$\vec{AD} = (1, 2, 1)$$

$$|m| = \sqrt{2^2 + 1^2} = \sqrt{5}$$

$$D' = D - \vec{AD}' = (0, 1, 2) - \left(\frac{6}{5}, 0, \frac{3}{5} \right) = \left(-\frac{6}{5}, 1, \frac{7}{5} \right)$$

$$A = \frac{D + D''}{2} \Rightarrow D'' = 2A - D = (-2, -3, 0)$$

$$D' = D + \vec{AD}' = (0, 1, 2) + \left(\frac{6}{5}, 0, \frac{3}{5} \right) = \left(\frac{6}{5}, 1, \frac{13}{5} \right)$$

$$\begin{aligned} D' &= A + \vec{AD}' = (-1, -1, 1) + \left(\frac{6}{5}, 0, \frac{3}{5} \right) \\ &= \left(\frac{6}{5} - 1, -1, \frac{3}{5} + 1 \right) \\ &= \left(\frac{1}{5}, -1, \frac{8}{5} \right) = \end{aligned}$$

③

⑧ orthogonal reflection of D in line AB.

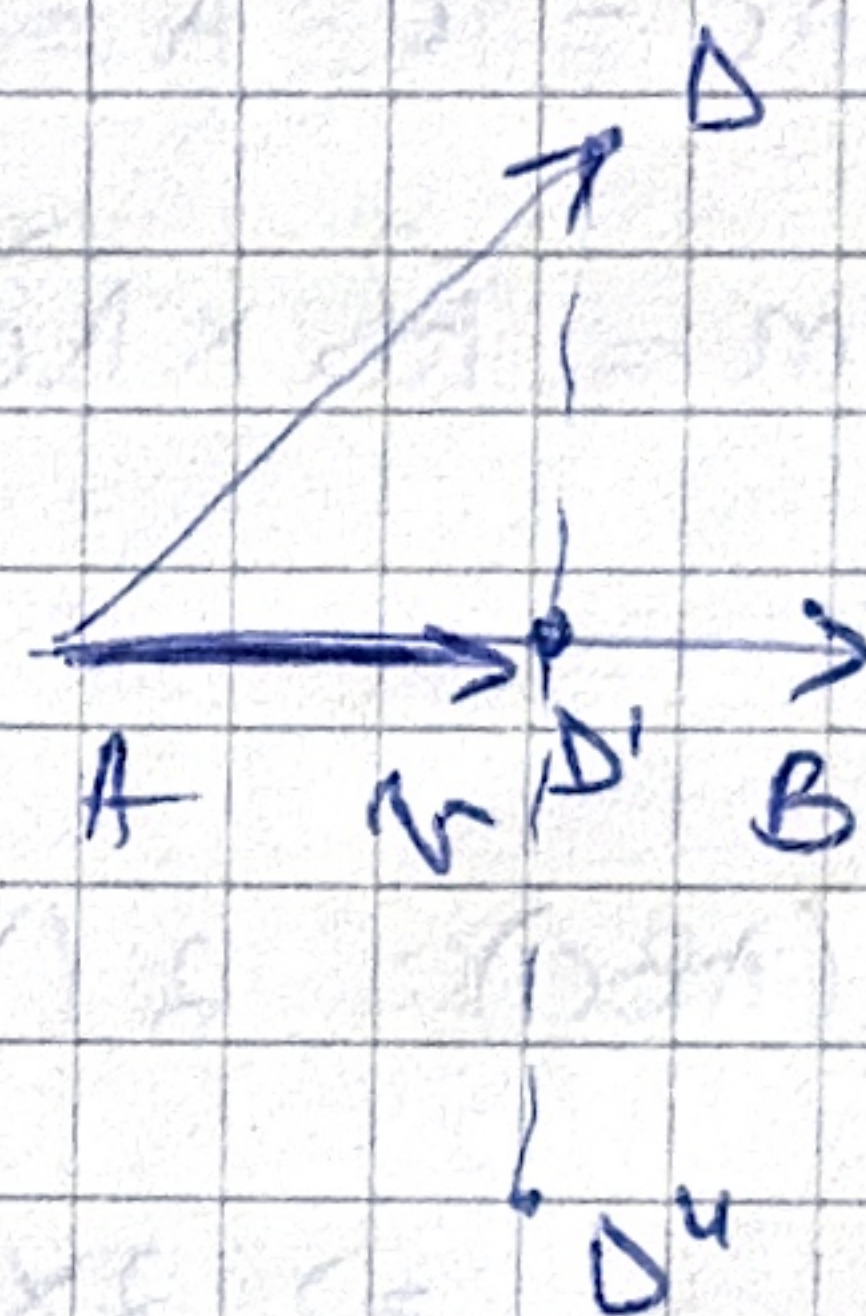
✓ $A(-1, -1, 1)$

$\vec{AB}(0, 2, 0) \Rightarrow v = (0, 2, 0)$ dir. vector.

$\vec{AD}(1, 2, 1)$

$P_{\vec{v}}^{\perp} \vec{AD} = \frac{\langle \vec{AD}, \vec{v} \rangle}{\langle \vec{v}, \vec{v} \rangle} \vec{v}$

$= \frac{4}{4} (0, 2, 0) = (0, 2, 0)$



$\langle \vec{AD}, \vec{v} \rangle = \langle (1, 2, 1), (0, 2, 0) \rangle = 0 + 4 = 4$

$\langle \vec{v}, \vec{v} \rangle = |\vec{v}|^2 = (\sqrt{2^2})^2 = 2^2 = 4$

$D' = A + P_{\vec{v}}^{\perp}(\vec{AD}) = (-1, -1, 1) + (0, 2, 0) = (-1, 1, 1)$

reflection: $D'' = 2D' - D$

$D' = \frac{D + D''}{2}$ (mid) $\nearrow$

$D'' = 2 \cdot (-1, 1, 1) - (0, 1, 2) = (-2, 2, 2) - (0, 1, 2)$
 $= (-2, 1, 0)$